

Diabetes Prediction using Machine Learning: Classification Model Comparison and Performance Evaluation.

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Course/Batch: Spring Internship Program 2026, Section-1

Institute: IDEAS – Institute of Data Engineering, Analytics and Science Foundation

Period of Internship: 21st January 2026 – 17th February 2026

Report submitted to:

IDEAS – Institute of Data Engineering, Analytics and Science Foundation, ISI Kolkata

1. Abstract

This project focuses on building machine learning classification models to predict whether a patient has diabetes using the Pima Indian Diabetes Dataset. The dataset contains medical attributes such as glucose level, blood pressure, BMI, insulin level, and age. The objective was to compare multiple classification algorithms and evaluate their performance using metrics such as accuracy, precision, recall, F1-score, and ROC-AUC score. Logistic Regression, K-Nearest Neighbors (KNN), and Support Vector Machine (SVM) models were implemented. The dataset was pre-processed and scaled before model training. The data was split into training and testing sets using stratified sampling. Model performance was evaluated on unseen test data. Comparative analysis was conducted to determine the best performing model. The results demonstrate the effectiveness of machine learning techniques in medical diagnosis prediction tasks.

2. Introduction

Diabetes is a chronic medical condition affecting millions of individuals worldwide. Early prediction and diagnosis can significantly reduce long-term health complications. With the advancement of data analytics and machine learning, predictive models can assist in identifying patients at risk.

This project applies supervised machine learning classification techniques to predict diabetes based on medical diagnostic measurements from the Pima Indian Diabetes Dataset.

Technologies and Tools Used:

- Python
- Google Colab
- Pandas and NumPy
- Matplotlib and Seaborn
- Scikit-learn

Topics covered during internship training:

- Data preprocessing
- Exploratory Data Analysis (EDA)
- Feature scaling
- Classification algorithms
- Model evaluation metrics
- ROC-AUC analysis

The project aims to demonstrate practical implementation of machine learning in healthcare prediction.

3. Project Objectives

1. To explore and analyze the diabetes dataset using descriptive statistics and visualization.
 2. To preprocess and scale the dataset appropriately for model training.
 3. To implement multiple classification algorithms.
 4. To evaluate models using accuracy, precision, recall, F1-score, and ROC-AUC.
 5. To compare model performances and identify the best performing classifier.
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4. Methodology

Step 1: Data Collection

The Pima Indian Diabetes Dataset was provided in CSV format. The dataset contains 768 records and 9 columns including 8 predictor variables and 1 target variable (Outcome).

Step 2: Data Inspection and Cleaning

- Checked dataset shape and structure.
 - Verified missing values.
 - Generated descriptive statistics using `.describe()`.
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Step 3: Feature and Target Definition

- Feature Matrix (X): All columns except 'Outcome'.
 - Target Vector (y): 'Outcome' column.
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Step 4: Train-Test Split

The dataset was divided into:

- 80% Training Data
- 20% Testing Data

Stratified sampling was applied to maintain class distribution.

Step 5: Feature Scaling

StandardScaler was applied to:

- Fit on training data.
 - Transform both training and testing datasets.
 - This ensures models like Logistic Regression, KNN, and SVM perform optimally.
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Step 6: Model Implementation

The following classification models were implemented:

1. Logistic Regression
2. K-Nearest Neighbors (n_neighbors = 5)
3. Support Vector Machine (kernel = linear)

Each model was trained on scaled training data and evaluated on scaled testing data.

Step 7: Model Evaluation

Evaluation Metrics Used:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC Score

Classification reports were generated for detailed performance analysis.

5. Data Analysis and Results

Descriptive Analysis

```
# Basic Libraries
import numpy as np
import pandas as pd

# Visualization
import matplotlib.pyplot as plt
import seaborn as sns

# Model Selection
from sklearn.model_selection import train_test_split

# Models
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.svm import SVC

# Evaluation
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, confusion_matrix, classification_report

# Ignore warnings
import warnings
warnings.filterwarnings("ignore")

# Load Dataset
df = pd.read_csv("diabetes (1).csv")

# Display first 5 rows
df.head()
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
0	6	148	72	35	0	33.6	0.627	50	1
1	1	85	66	29	0	26.6	0.351	31	0
2	8	183	64	0	0	23.3	0.672	32	1
3	1	89	66	23	94	28.1	0.167	21	0
4	0	137	40	35	168	43.1	2.288	33	1

<pre> # If df is not defined earlier, load it again safely try: df except NameError: df = pd.read_csv("diabetes (1).csv") # Descriptive statistics df.describe() </pre>									
...	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
count	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000
mean	3.845052	120.894531	69.105469	20.536458	79.799479	31.992578	0.471876	33.240885	0.348958
std	3.369578	31.972618	19.355807	15.952218	115.244002	7.884160	0.331329	11.760232	0.476951
min	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.078000	21.000000	0.000000
25%	1.000000	99.000000	62.000000	0.000000	0.000000	27.300000	0.243750	24.000000	0.000000
50%	3.000000	117.000000	72.000000	23.000000	30.500000	32.000000	0.372500	29.000000	0.000000
75%	6.000000	140.250000	80.000000	32.000000	127.250000	36.600000	0.626250	41.000000	1.000000
max	17.000000	199.000000	122.000000	99.000000	846.000000	67.100000	2.420000	81.000000	1.000000

The dataset contains 768 samples and 8 predictor variables. Statistical summaries indicate variation in glucose levels, BMI, and age among patients.

Model Evaluation Results

<pre> from sklearn.linear_model import LogisticRegression from sklearn.metrics import classification_report lr = LogisticRegression(max_iter=1000) lr.fit(X_train_scaled, y_train) y_pred_lr = lr.predict(X_test_scaled) print(classification_report(y_test, y_pred_lr)) </pre>					
...	precision	recall	f1-score	support	
0	0.76	0.82	0.79	100	
1	0.61	0.52	0.56	54	
accuracy			0.71	154	
macro avg	0.68	0.67	0.67	154	
weighted avg	0.71	0.71	0.71	154	

```

from sklearn.metrics import roc_auc_score

# Get probability scores instead of class predictions
lr_probs = lr.predict_proba(X_test_scaled)[i, 1]
knn_probs = knn.predict_proba(X_test_scaled)[i, 1]

# Recreate SVM with probability enabled
svm = SVC(kernel='linear', random_state=42, probability=True)
svm.fit(X_train_scaled, y_train)
svm_probs = svm.predict_proba(X_test_scaled)[i, 1]

# Calculate ROC-AUC using probabilities
roc_lr = roc_auc_score(y_test, lr_probs)
roc_knn = roc_auc_score(y_test, knn_probs)
roc_svm = roc_auc_score(y_test, svm_probs)

# Store results
model_scores = {
    'LogisticRegression': roc_lr,
    'KNN': roc_knn,
    'SVM': roc_svm
}

print(model_scores)

# Identify best model
best_model = max(model_scores, key=model_scores.get)
print("Best Model:", best_model)

{'LogisticRegression': np.float64(0.8229629629629629), 'KNN': np.float64(0.7404629629629), 'SVM': np.float64(0.8280555555555555)}
Best Model: SVM

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Model	ROC-AUC Score
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Logistic Regression:	0.8229
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KNN:	0.7404
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SVM:	0.8280
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Best Model:	SVM
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Based on ROC-AUC comparison, the best performing model was:

Support Vector Machine (SVM) with a ROC-AUC score of 0.8280.

The ROC-AUC metric measures a model's ability to distinguish between diabetic and non-diabetic patients. A higher ROC-AUC score indicates better classification performance. Among the evaluated models, SVM demonstrated slightly superior discriminatory power.

6. Conclusion

This project successfully demonstrated the implementation of classification algorithms for diabetes prediction. The dataset was preprocessed and scaled to ensure optimal model performance. Logistic Regression, KNN, and SVM models were compared using multiple evaluation metrics. Among the tested models, the best performing classifier was identified based on ROC-AUC score. The results highlight the effectiveness of machine learning techniques in medical diagnosis prediction tasks. Future improvements may include hyperparameter tuning, cross-validation, and ensemble methods to further enhance predictive accuracy.

7. APPENDICES

1. References

- Scikit-learn Documentation
- Pandas Documentation
- Pima Indian Diabetes Dataset

2. GitHub Repository Link

<https://github.com/Prithwi1234-wq/Spring-Internship-2026-Diabetes-Prediction>

3. Code Repository

All implementation code is available in the GitHub repository submitted along with this report.